



## Contact

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## Hard Skills

### Machine learning and deep learning

- PyTorch, Tensorflow
- Model optimization: quantization, compression, ONNX, TFLite

### Audio & signal processing

- Musical acoustics, psychoacoustics, spatial audio
- Audio augmentation, feature extraction

### MLOps & development

- Production deployment & embedded systems optimization
- Model evaluation and custom metrics
- CI/CD pipelines

### Programming tools

- Python (signal and audio libraries)
- C++, Matlab, Faust
- Git, Linux, cloud computing (AWS)

### Music

- Piano (+25 years), music theory
- DAWs, synthesizers, sound design, mixing, mastering

## Soft Skills

- Technical leadership | Autonomous | Problem-solver
- Cross-functional collaboration | Rigorous | Strategic thinking

## Education

|             |                                                                                                                                      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 2018 – 2021 | <b>PhD degree</b><br>GIPSA-lab, Grenoble                                                                                             |
| 2017 – 2018 | <b>Research master degree</b><br><i>Acoustics, audio signal processing and computer music (ATIAM)</i><br>Sorbonne Universités, Paris |
| 2013 – 2017 | <b>Engineering degree</b><br><i>Computer science</i><br>Centrale Lille, Lille                                                        |
| 2011 – 2013 | <b>Preparatory classes</b><br>Lycée Marcelin Berthelot, Saint-Maur                                                                   |

## Hobbies

Volleyball (national level competitions), music making, science & history books, museums, chess, nature exploration, hiking,

## About Me

Audio ML Engineer with a strong research foundation (PhD, postdoc) and practical experience shipping deep learning models for real-time audio applications on embedded devices. I'm driven by the convergence of AI and audio/music technology, and I thrive in environments where I can push both research boundaries and production quality. My 25+ years as a pianist and my work in music production fuel my passion for this field.

## Professional Experience

### Sonaid | Audio ML engineer

2024 – present

- Built end-to-end deep learning pipeline from scratch: dataset creation with custom annotations and audio augmentation strategies, model architecture, training, and evaluation with custom metrics
- Researched and implemented state-of-the-art audio processing techniques for audio event detection
- Optimized models for embedded systems through compression, quantization, latency reduction, and memory optimization while balancing accuracy vs. computational constraints
- Prepared and deployed models for production environments
- Led technical strategy: defined AI roadmap, prioritized features, and integrated customer feedback into model improvements
- Collaborated cross-functionally with product, hardware, and backend teams while maintaining technical documentation

### LS2N, Centrale Nantes | Postdoctoral researcher

2022 – 2023

Bandwidth extension of musical signals with differentiable models

- State-of-the-art review of bandwidth extension methods
- Design and evaluation of several DDSP models for monophonic and polyphonic bandwidth extension

### Orange Labs & GIPSA-lab | PhD researcher

2018 – 2021

Deep learning for speaker counting and localization with Ambisonics signals

- State-of-the-art review of sound source counting and localization
- Preparation of datasets including multichannel multi-source speech signals with reverberation and noise in Ambisonics format
- Deep learning models for speaker counting, speaker localization and hybrid methods : CNN, RNN, CRNN, ResNet, self-attention
- Investigations regarding a novel 3D audio representation

### IRCAM | Research intern

Feb. – Jul. 2018

Automatic drums transcription with deep learning

### Audionamix | Research intern

Apr. – Aug. 2017

Audio-to-lyrics alignment for polyphonic music

### CCRMA & Mines ParisTech | Research intern

Jun. – Aug. 2016

Physical modeling based sound synthesis with Faust